Unfreezing HDDs after Purge Interrupt (YEC)

Why it happens:

Hard Disk Drives (HDD) that are interrupted in the middle of their purge/sanitize cycle will be frozen and on standby until they are allowed to complete their cycle. Modern SSDs have safeguards against this type of behavior generally.

What is actually happening:

When issuing a Purge command, the operator is using a low-level firmware command which is sent directly to the drive. The drive uses code baked into the firmware to erase all sectors from the inside-out, as opposed to an operating system wiping the drives from the outside-in.

During the purge process, the drive prevents itself from receiving any other inputs until completion.

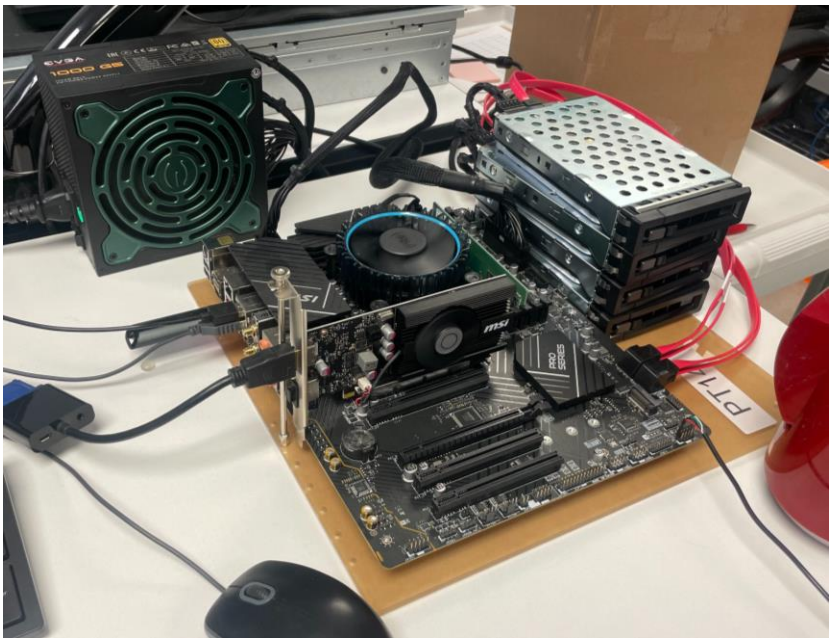
If the wiping machine (YEC) freezes mid-purge, loses power, or otherwise, the HDD's may:

- ➢ Remain in a **"sanitizing" or "frozen"** state
- ➢ Reject normal read/write I/O
- ➢ Hide or restrict some commands until sanitize completes/aborts

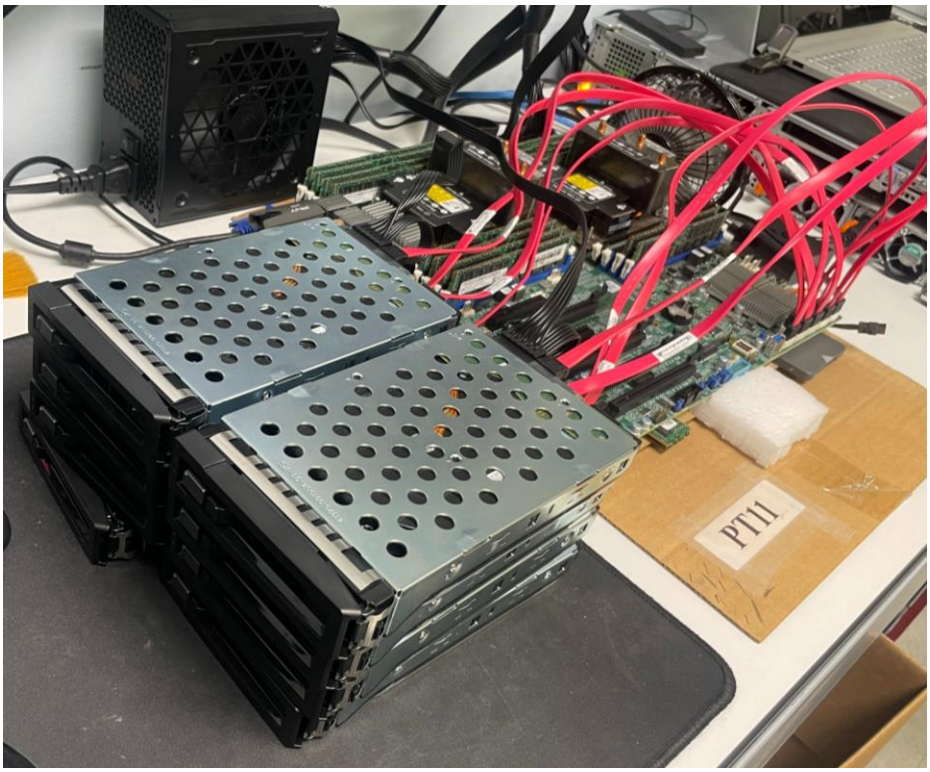
How to resolve:

In order to resolve the problem the drives must be connected to a low-level controller or board capable of sending commands directly to the drives via the SystemRescue operating system.

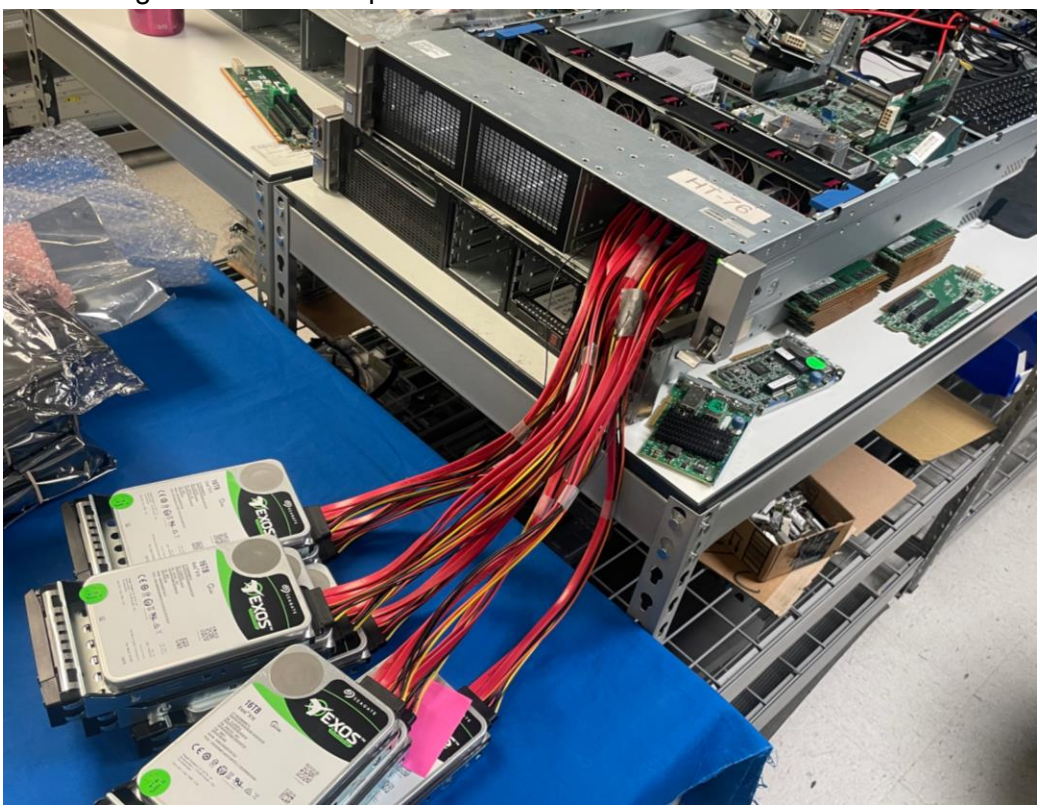
1. Connect to a PC motherboard via SATA to issue commands



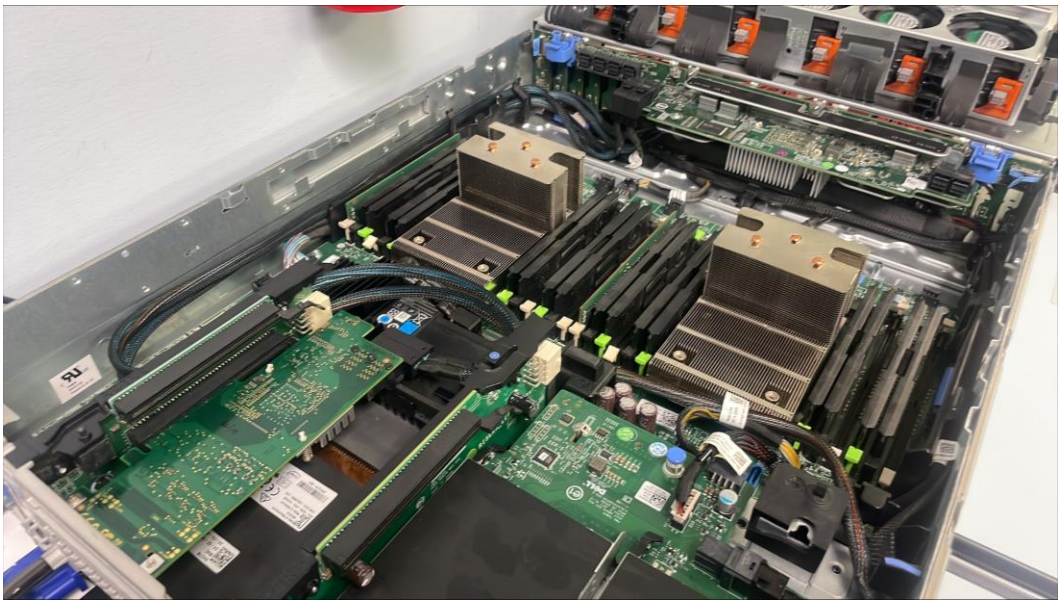
2. Connect to a server motherboard via SATA to issue commands.



3. Connecting to a server backplane to issue commands.



4. Connect a Storage Controller in Initiator Target (IT) mode via PCIE slot, connected to a backplane/expander to issue commands.



★ Note: Whichever method you choose, be aware that the purge/sanitize process will take a while and prevent you from using that system for other purposes.

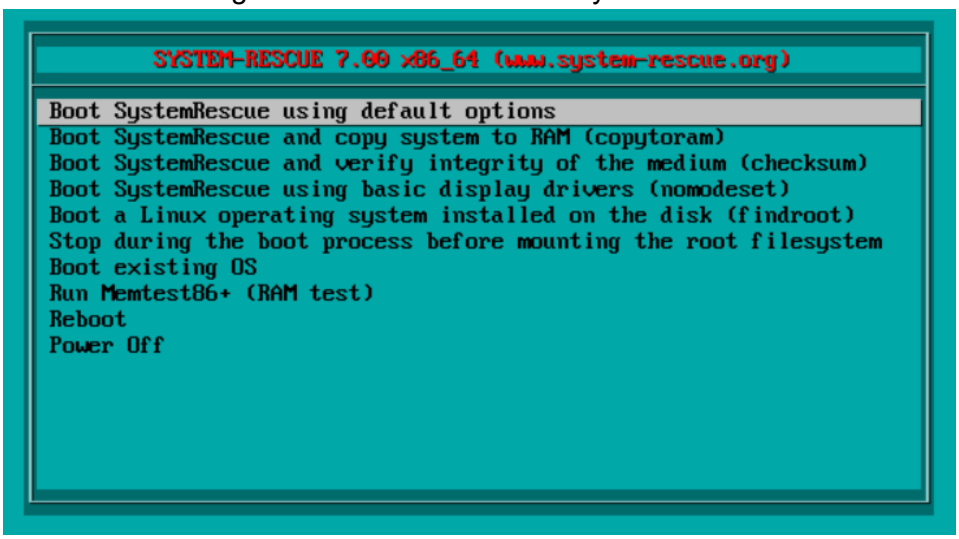
★ The 16TB/12Gbps SATA HDD drive took around 7-10 days to complete.

System Rescue

https://www.system-rescue.org/Download/

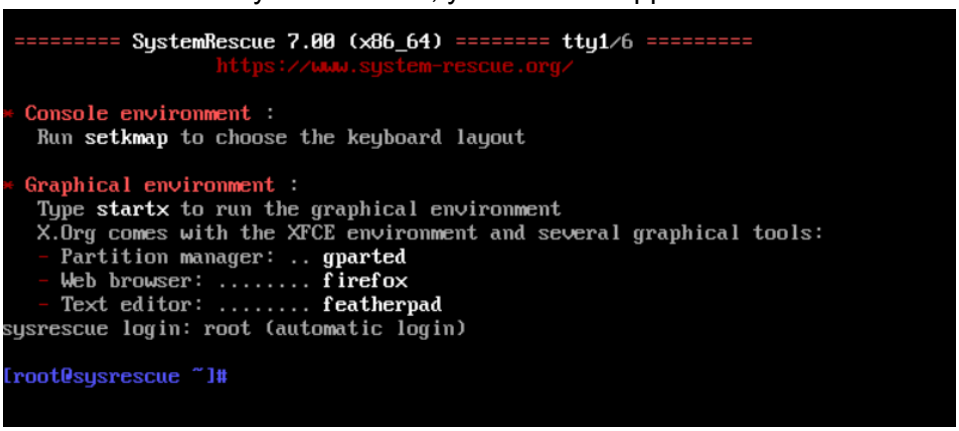
How to Use:

1. Download the live boot ISO to USB (systemrescue-12.02-amd64.iso)
2. Turn on the testing station and boot into the systemrescue ISO from the boot menu.

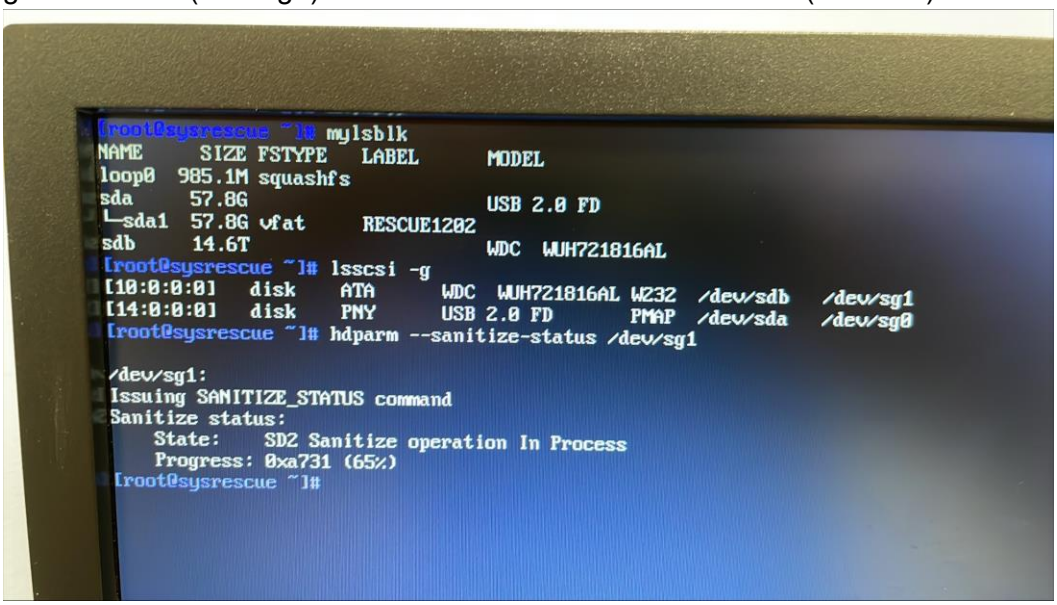


➢ Note: It may drop you into a black and yellow shell, type “exit” to proceed

3. Choose default, or copy the system to RAM if you would like to remove the USB stick.
4. Once loaded into SystemRescue, you will be dropped into a command line shell.



5. You can list all block devices with the built in alias mylsblk or lsscsi -g to show their generic name (/dev/sg1) associated with their standard name (/dev/sdb)



6. You can query the sanitize status with hdparm --sanitize-status /dev/sgX (or /dev/sdX)
7. Allow the drives to run until the cycle is complete. Once finished, the drives should be back to normal operation.

Note: The drives may spin-up upon power connection, and may enumerate with lsblk briefly, followed by spinning down and returning to standby. I believe this may indicate a failing drive sensor that prevents the drive from sending or receiving input/output in which the manufacturer may need to be contacted in order to be restored.

Note: If the drives do not enumerate on a normal motherboard or server, a storage controller in IT mode may be required to interface with the drives at their lowest level.

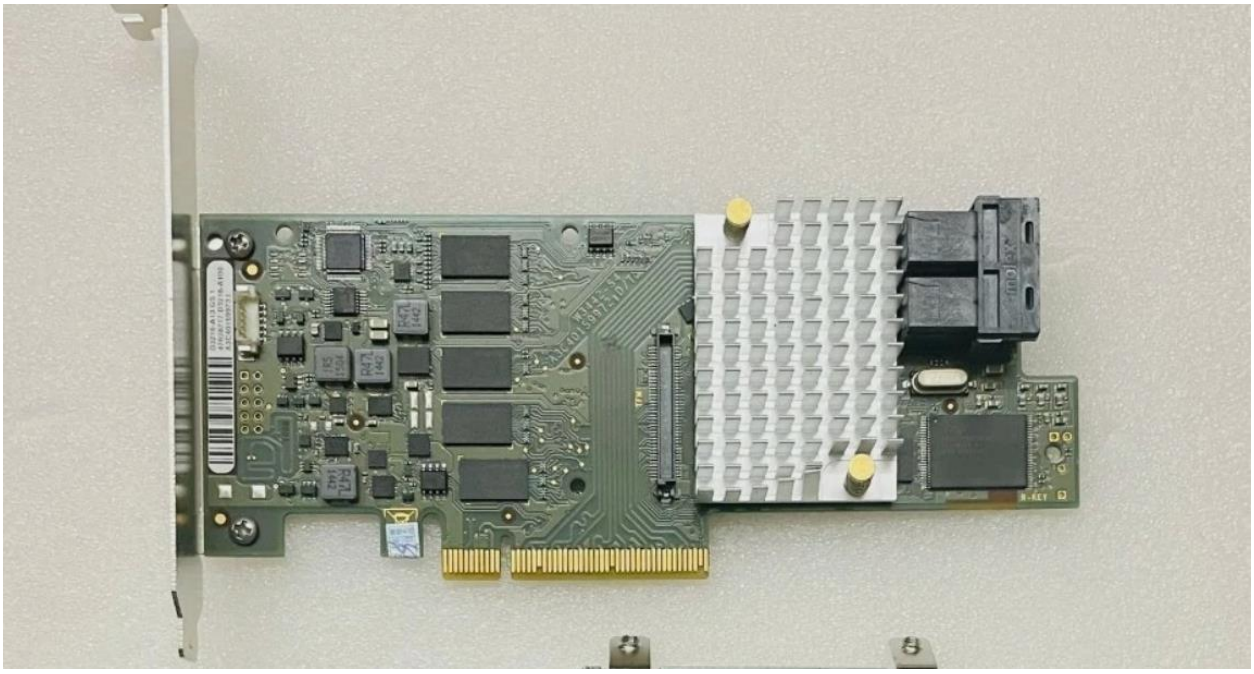
Other Useful Commands:

1. sudo hdparm -I /dev/sdX # Full identify for HDD/SSD
2. sudo sg_turs --progress /dev/sgX # Check testing unit ready status
3. sudo sg_readcap -l /dev/sgX # Check drive capacity & sector size
4. sudo smartctl -a /dev/sdX # Check drive health S.M.A.R.T information
5. sudo sg_sanitize --fail /dev/sgX # Pre-emptively exit sanitize mode
6. sudo sg_format -F -s 512 -f 0 -v /dev/sgX # Reformat drive, sector 512, info protect off

- **smartctl** → Drive health, SMART data, and unit tests (ALL drive types)
- **hdparm** → SATA/ATA disk information (mostly for SSD/HDD)
- **sg3_utils** → SCSI/SAS/SATA *low-level* control, including sanitize/format, particularly for enterprise SAS/SATA and some USB

Broadcom LSI 9300-8i SAS3008 PCI-Express Fusion-MPT SAS-3 (rev 02)

FUJITSU D3307-A12 PRAID CP400i



Card Description:

The Broadcom / LSI SAS 9300-8i is a 12 Gb/s SAS host bus adapter (HBA) based on the SAS3008 controller. It's designed to provide high-bandwidth, low-level passthrough access to SAS and SATA drives **without RAID abstraction** when run in IT (initiator-target) mode.

- ➢ Fujitsu OEM version of the Broadcom 9300-8i, flashed with latest IT-Mode firmware.
- ➢ Compatible with FreeNAS, TrueNAS, Unraid, Windows 10, Windows Server 2008, 2012, 2016, 2019, and Linux.
- ➢ 12GBps SAS / SATA
- ➢ LSI SAS3008 chipset
- ➢ Controller Version 15.00.00.00
- ➢ IT-Mode Firmware 13.0.0.0
- ➢ 2 ports miniSAS SFF-8643

Card Specs:

- Lifecycle: Active
- Host Bus Type: x8 lane PCI Express® 3.0
- Physical Dimensions: Low profile (2.6" x 6.0")
- Devices Supported: 1024 Non-RAID SAS/SATA devices
- Fusion MPT Support: Fusion MPT™ 2.5
- Internal Connectors: 2 Mini-SAS HD SFF8643
- Internal Ports: 8
- Operating Temperature: 0°C to 55°C
- Operating Voltage (V): +12V +/-8%, 3.3V +/-9%
- OS Support: Microsoft Windows, Linux (SuSE , Red Hat), Solaris, VMware , FreeBSD See the downloads page for details on versions The HBA supports Solaris 10 and Oracle® provides a built in driver; installation of an LSI driver is not required.
- PCI Power: 13W
- Cable Support: Passive Copper
- I/O Controller: LSI SAS3008
- MTBF: > 2,800,000 hours
- Operating Humidity: 5% to 90% Non-condensing
- Brackets: Full height and low profile vented
- Regulatory Certifications: Australia/New Zealand (AS/NZS 3548), Canada (ICES-003), EMC: Class B-US (CFR 47, P15B), Europe (EN55022/EN55024), Japan (V-3/02.04), Korea (N-B 2-0001-404), RoHS, Safety: EN60950, WEEE
- Data Transfer Rates: 12Gb/s SAS-3 Compliant

Card Installation:

1. **Open server/PC and Install the 9300-8i** in a PCIe slot.
2. **Attach cables** from card to server backplane (A0/B0) or attach mini-sas hd connectors directly to drives along with power cables.
3. **Boot into OS (Linux/SystemRescue**):

Note: if the storage controller was not flashed with BIOS capability, you will not be able to load the OS drive while connected to it. In that case, you should connect the OS drive to an internal SATA/USB or Rear-cage and load the operating system independently from the controller.

!! Unplug any other storage controllers from the backplane, you should only have one controller connected otherwise corruption could occur to drives !!

Flashing BIOS/Firmware:

- https://github.com/EverLand1/9300-8i_IT-Mode/tree/main/Files
- Flashing-firmware-and-bios-on-lsi-sas-hbas
- Ebay Listing Purchased From